

Summary and recommendations

1. Objective

The objective of this analysis is to **identify key factors influencing customer churn** and understand patterns in customer behavior using exploratory data analysis (EDA). The goal is to provide **actionable insights to improve customer retention**.

2. Data Overview

- Dataset includes customer demographics, service usage, and churn status.
 - Overall churn rate is approximately **26.5%**, meaning **about 1 in 4 customers leave the service**.
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3. Key Findings

a) Demographic Insights

- **Senior Citizens:** ~**41% churn rate**, compared to ~**24% for non-seniors**
→ Indicates **higher vulnerability among older customers**
 - **Gender:** No significant difference in churn
→ **Not a major influencing factor**
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b) Service-Based Insights

- **Internet Service**
 - Fiber Optic: ~**41–42% churn** (highest)
 - DSL: ~**19–20% churn**
 - No Internet: ~**7–8% churn**
→ **Fiber users are most likely to churn**
- **Online Security**
 - Without: ~**41–42% churn**
 - With: ~**14–15% churn**
→ **Almost 3× higher churn without security**
- **Tech Support**
 - Without: ~**41–42% churn**
 - With: ~**15% churn**
→ Strong impact on **customer retention**
- **Device Protection**
 - Without: ~**39–40% churn**

- With: **~22–23% churn**
 - **Online Backup**
 - Without: **~39–40% churn**
 - With: **~21–22% churn**
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c) Entertainment Services

- **Streaming TV & Movies**
 - Churn difference is smaller (**~30–33% vs ~26–28%**)
→ **Less influence on churn decisions**
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4. Insights & Interpretation

- Customers without **support and security services** are **2–3 times more likely to churn**
 - **Fiber optic users** show the highest dissatisfaction or price sensitivity
 - **Value-added services** (security, support, protection) are key drivers of retention
 - **Basic and entertainment services** have comparatively **lower impact**
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5. Conclusion

Customer churn is primarily driven by **lack of support, security, and service reliability** rather than demographic factors or entertainment features.

👉 To reduce churn, the company should:

- Promote **bundled plans with tech support and security services**
- Improve **fiber optic service quality and pricing strategy**
- Focus on **high-risk segments like senior citizens and unsupported users**

Final Insight:

Enhancing **customer support and perceived value** can significantly reduce churn from **~40% levels to nearly 15–20%**, leading to stronger customer retention and long-term growth.